MATHEMATICS
Introduction to Modern Algebra II
Quiz 5.

1) What is the degree and the irreducible polynomial of $\sqrt{2}+\sqrt{5}$ over $\mathbb{Q}$?

Proof. We have $(\sqrt{2}+\sqrt{5})^2 = 7+2\sqrt{10}$, $(\sqrt{2}+\sqrt{5})^4 = 89+28\sqrt{10}$. If $\alpha = \sqrt{2}+\sqrt{5}$, then $\alpha^4 - 14\alpha^2 + 9 = 0$. We claim that $P(x) = x^4 - 14x + 9$ is the irreducible polynomial of α. The only possible roots are $1, -1, 3, -3$ and so $P(x)$ has no roots and hence no linear factors. Suppose that it has a quadratic term and write

$$x^4-14x+9 = (x^2+ax+b)(x^2+cx+d) = x^4+(a+c)x^3+(b+ac+d)x^2+(da+cb)x+bd.$$

Then $c = -a$. But then $0 = da + cb = a(d-b)$ so that $a = 0$ or $d = b$. Also $bd = 9$ and so $(b,d) \in \{(1,9),(3,3),(9,1),(-1,-9),(-3,-3),(-9,-1)\}$ so that $b+d \in \{10,6,-6,-10\}$. If $a = 0$, then $-14 = b+d =$, a contradiction. If $a \neq 0$, then $d = b$ so $b+d = \pm 6$ and $-14 = b+d-ac = \pm 6 - a^2$ i.e. $a^2 = \pm 6 + 14$, a contradiction. Therefore $P(x) = x^4 - 14x + 9$ is irreducible and $[\mathbb{Q}(\sqrt{2}+\sqrt{5}) : \mathbb{Q}] = 4$.

Alternative and easier proof that $[\mathbb{Q}(\sqrt{2}+\sqrt{5}) : \mathbb{Q}] \neq 2$**.** If this where not the case then $1, \alpha, \alpha^2$ would be linearly dependent and hence so would $1, \alpha, \sqrt{10}$. So we write $a + b(\sqrt{2}+\sqrt{5}) + c\sqrt{10} = 0$ so that

$$a^2 + 10c^2 + 2ac\sqrt{10} = (a + c\sqrt{10})2 = (b(\sqrt{2}+\sqrt{5}))^2 = b^2(7+\sqrt{10}).$$

Thus $a^2 + 10c^2 = 7b^2 = 14ac$, so that $\frac{a}{c} = -7 \pm \sqrt{49-10} \notin \mathbb{Q}$, a contradiction. ☐

2) Show that if a, b are algebraic over F, then $[F(a,b) : F] \leq [F(a) : F][F(b) : F]$ **and** show that if $[F(a) : F]$ and $[F(b) : F]$ are coprime, then $[F(a,b) : F] = [F(a) : F][F(b) : F]$.

Proof. Note that $[F(a,b) : F(a)] \leq [F(b) : F]$ (if $P(x) \in F[x]$ is the irreducible polynomial of b over F, then as $P(x) \in F(a)[x]$, the irreducible polynomial of b over $F(a)$ divides $P(x)$). We thus have

$$[F(a,b) : F] = [F(a,b) : F(a)][F(a) : F] \leq [F(a) : F][F(b) : F].$$

Note that $[F(a) : F]$ divides $[F(a,b) : F]$ (and so does $[F(b) : F]$). Thus if $[F(a) : F]$ and $[F(b) : F]$ are coprime, then their product divides $[F(a,b) : F]$. As $[F(a,b) : F] \leq [F(a) : F][F(b) : F]$ we obtain $[F(a,b) : F] = [F(a) : F][F(b) : F]$.

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